Moving Robots

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1726 | Mathematics | https://cses.fi/problemset/task/1726

Each square of an 8 × 8 chessboard has a robot. Each robot independently moves k steps, and there can be many robots on the same square.

On each turn, a robot moves one step left, right, up or down, but not outside the board. It randomly chooses a direction among those where it can move.

Your task is to calculate the expected number of *empty* squares after k turns.

Input

The only input line has an integer k.

Output

Print the expected number of empty squares rounded to six decimal places (rounding half to even).

Constraints

- $1 \le k \le 100$

Example

Input
10

Output
23.120740